

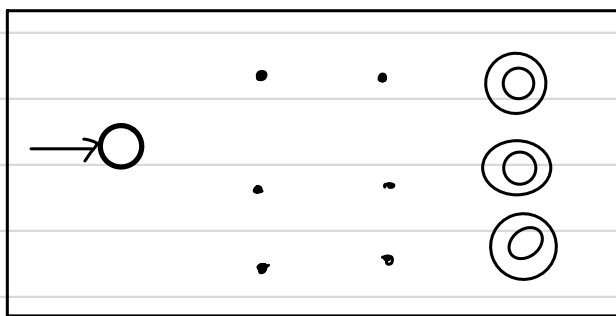
$$AB = \{xy \mid x \in A \text{ and } y \in B\}$$

If A and B are regular then AB is regular

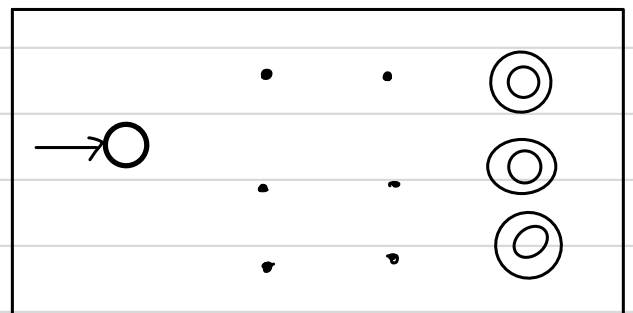
$\exists$ DFA M_1 s.t. $L(M_1) = A$ and $\exists$ DFA M_2 s.t. $L(M_2) = B$

To construct N s.t. $L(N) = AB$
 $\hookrightarrow$ on ϵ -NFA.

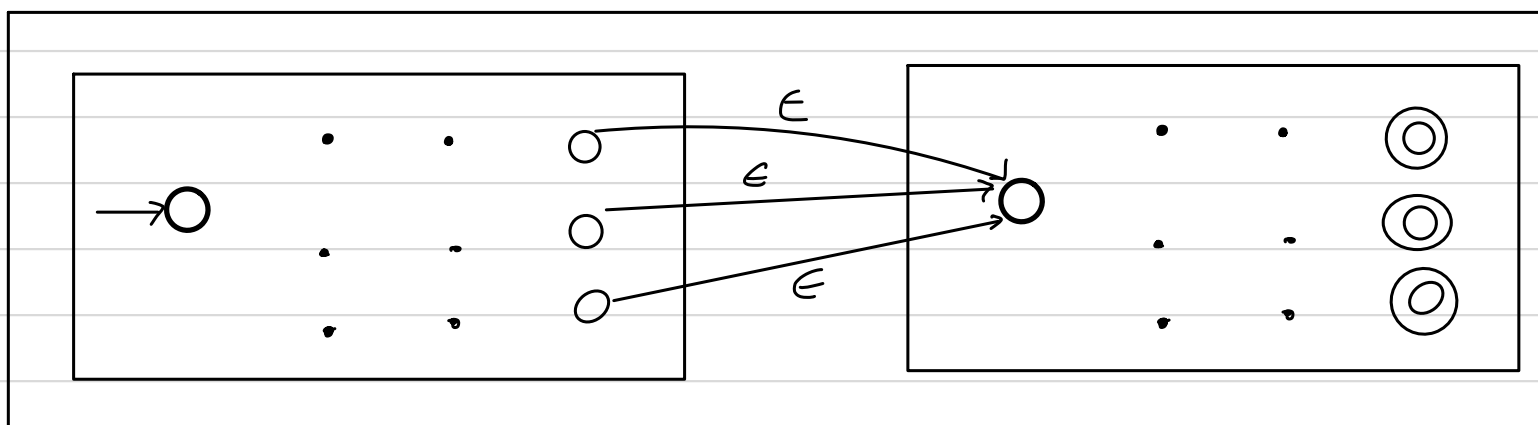
M_1 (DFA)



M_2 (DFA)



NFA: N



$$L(N) = AB$$

$$M_1 = (Q_1, \Sigma, \Delta_1, S_1, F_1) \quad M_2 = (Q_2, \Sigma, \Delta_2, S_2, F_2)$$

$$L(M_1) = A \quad L(M_2) = B$$

$$N = (Q, \Sigma, \Delta, S, F) \quad \text{st} \quad L(N) = AB$$

$$Q = Q_1 \cup Q_2 \quad S = S_1 \quad \text{and} \quad F = F_2$$

Defining the transition function Δ

$$\Delta(q, a) = \begin{cases} \Delta_1(q, a) & q \in Q_1 \text{ and } q \notin F_1 \\ \Delta_1(q, a) & q \in F_1 \text{ and } a \neq \epsilon \\ \Delta_1(q, a) \cup S_2 & q \in F_1 \text{ and } a = \epsilon \\ \Delta_2(q, a) & q \in Q_2. \end{cases}$$

Let $A \subseteq \Sigma^*$

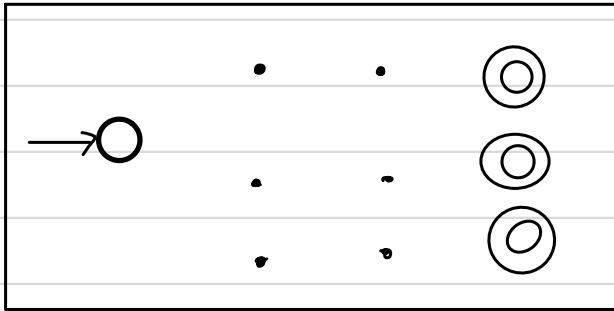
$$A^* = \{x_1 x_2 \dots x_n \mid n \geq 0 \text{ and } x_i \in A, 1 \leq i \leq n\}.$$

$$= \{\epsilon\} \cup A \cup A^2 \cup \dots$$

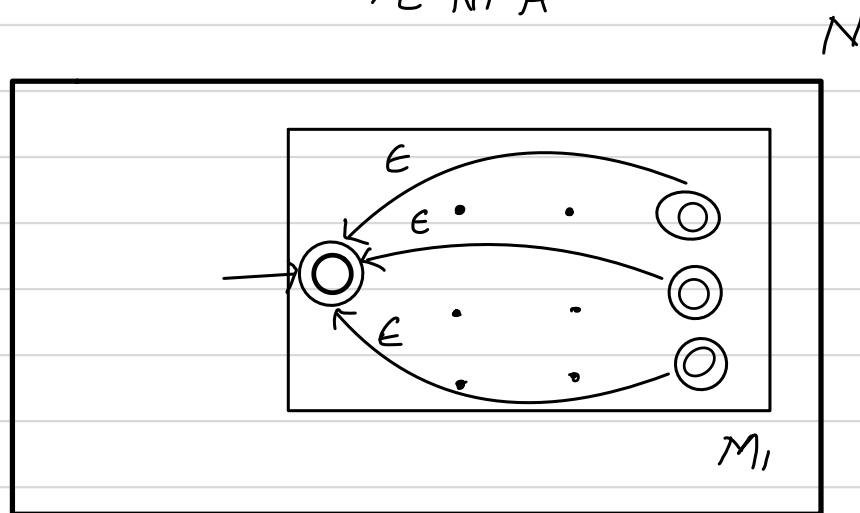
Statement. if A is regular then A^* is regular

$\exists$ a DFA M_1 s.t. $L(M_1) = A$.

M_1 (DFA)



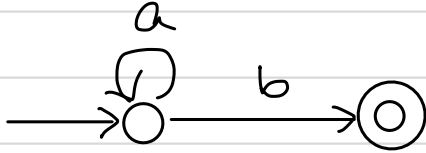
To construct an NFA N s.t. $L(N) = A^*$
 $\hookrightarrow$ E-NFA



$$L(N) = A^*.$$

Example

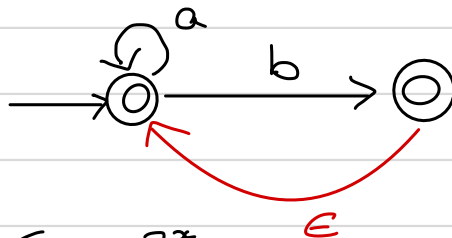
M:



$$L(M) = \{a^n b \mid n \geq 0\} = A.$$

$$A^* = \{\epsilon\} \cup \{\text{strings ending with } b\}.$$

$$N \text{ s.t. } L(N) = \Sigma^*$$



$$L(N) = \{a, b\}^*$$

$$\text{Thus } L(N) \neq A^*$$

Let $A \subseteq \Sigma^*$

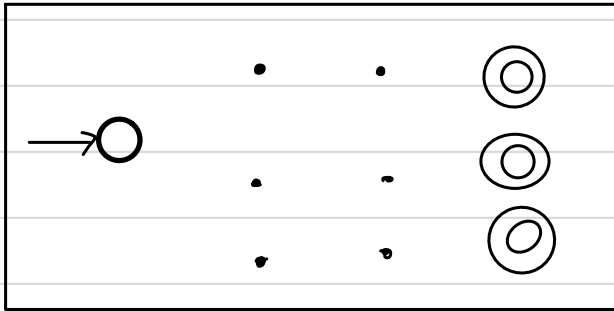
$$A^* = \{x_1 x_2 \dots x_n \mid n \geq 0 \text{ and } x_i \in A, 1 \leq i \leq n\}.$$

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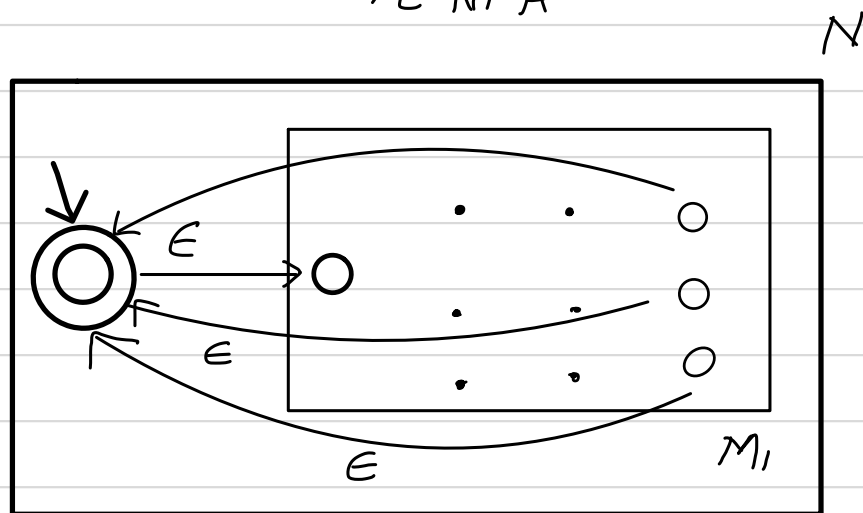
Statement. if A is regular then A^* is regular

$\exists$ a DFA M_1 s.t. $L(M_1) = A$.

M_1 (DFA)



To construct an NFA N s.t. $L(N) = A^*$
 $\hookrightarrow$ E-NFA



$$L(N) = A^*.$$

Suppose $A \subseteq \Sigma^*$ is regular then

$$\exists M_1 = (Q_1, \Sigma, \Delta_1, S_1, F_1) \text{ s.t. } L(M_1) = A$$

Construct $N = (Q, \Sigma, \Delta, S, F)$ s.t. $L(N) = A^*$

$$Q = Q_1 \cup \{s_0\} \quad S = \{s_0\} \quad F = \{s_0\}$$

Transition function Δ

$$\Delta(q, a) = \begin{cases} \Delta_1(q, a) & \text{if } q \in Q_1 \text{ and } q \notin F_1 \\ \Delta_1(q, a) & \text{if } q \in F_1 \text{ and } a \neq \epsilon \\ \Delta_1(q, a) \cup \{s_0\} & \text{if } q \in F_1 \text{ and } a = \epsilon \\ S_1 & \text{if } q = s_0 \text{ and } a = \epsilon \\ \emptyset & \text{if } q = s_0 \text{ and } a \neq \epsilon. \end{cases}$$

$$\text{DFA} = (M, \Sigma, S, \delta, F) \quad \text{NFA} = (M, \Sigma, \Delta, S, F)$$

- $A \subseteq \Sigma^*$ is regular if $\exists$ DFA M s.t. $L(M) = A$.

- For every NFA N , $\exists$ DFA M s.t. $L(M) = L(N)$
(Subset Construction)

- $A \subseteq \Sigma^*$ is regular if $\exists$ NFA N s.t. $L(N) = A$.

Pattern Matching - Classify Strings.

> Is *.pdf - represents files with pdf extension.
Pattern γ

$$L(\gamma) = \{x \mid x \text{ matches } \gamma\}$$

> Is ab*.pdf
Atomic pattern Compound pattern.

Σ - alphabet set

Atomic Patterns.

Syntax

Semantics : $L(\gamma) \subseteq \Sigma^*$.

$a \in \Sigma$

$$L(a) = \{a\}$$

ϵ

$$L(\epsilon) = \{\epsilon\}.$$

$\emptyset$

$$L(\emptyset) = \emptyset \quad - \text{matches nothing.}$$

$\#$

$$L(\#) = \Sigma$$

@

$$L(@) = \Sigma^*$$

Compound Patterns.

Syntax.

Semantics

$\alpha + \beta$

$$L(\alpha + \beta) = L(\alpha) \cup L(\beta)$$

$\alpha \cap \beta$

$$L(\alpha \cap \beta) = L(\alpha) \cap L(\beta)$$

$\alpha \beta$

$$L(\alpha \beta) = L(\alpha) L(\beta) \\ = \{xy \mid x \in L(\alpha) \text{ and } y \in L(\beta)\}.$$

α^*

$$L(\alpha^*) = L(\alpha)^0 \cup L(\alpha)^1 \cup L(\alpha)^2 \cup \dots \\ = \{x_1 x_2 \dots x_n \mid n \geq 0; x_i \in L(\alpha), 1 \leq i \leq n\}$$

α^+

$$L(\alpha^+) = L(\alpha)^1 \cup L(\alpha)^2 \cup \dots \\ = L(\alpha)^+$$

$\neg \alpha$

$$L(\neg \alpha) = \overline{L(\alpha)} = \Sigma^* - L(\alpha)$$

Examples.

$$\Sigma^* = L(\underbrace{@}_{\text{atomic}}) = L(\underbrace{\#^*}_{\text{compound}}).$$

$$\{a\} = L(a)$$

- $\alpha = @a @ a @ a @$ $L(\alpha) = \{x \in \Sigma^* \mid x \text{ has at least 3 occurrences of } a\}.$
- $\alpha = @a @ b @$ - $x a y b z$ for $x, y, z \in \Sigma^*.$
- Strings with no occurrence of the symbol a .

$$(\# \cap \neg a)^*$$